

Adaptive rough fuzzy conditional spatial clustering algorithms for noisy brain tissue segmentation

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HIGHLIGHTS

- Developed adaptive rough-fuzzy clustering algorithms for noisy brain MRI tissue segmentation.
- Introduced rough set-based cluster centroid selection to improve clustering convergence.
- Proposed algorithms leverage spatial information to effectively handle overlapping & ambiguous brain image data.
- Demonstrated superior performance in robustly segmenting brain tissues under varying noise conditions.
- Validated techniques on multiple datasets, showing high accuracy and robustness against noise & inhomogeneity.

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ABSTRACT

Medical image segmentation plays a critical role in clinical diagnosis and analysis, particularly for brain MRI, where tissue boundaries are often ambiguous due to noise, intensity inhomogeneity, and overlapping anatomical structures. In this work, we introduce a set of *adaptive rough-fuzzy conditional spatial clustering algorithms* designed to improve brain tissue segmentation under such conditions. Our approach integrates rough set theory for handling uncertainty with fuzzy membership functions to capture gradual region transitions, while a novel rough set-based centroid initialization strategy reduces randomness and accelerates algorithm convergence. Unlike conventional clustering techniques, our method adaptively incorporates spatial context to enhance robustness against noise and preserves local anatomical consistency. Comprehensive experiments conducted on multiple benchmark brain MRI datasets demonstrate statistically significantly better segmentation accuracy than existing methods.

1. Introduction

The role of image segmentation in medical diagnosis is unparalleled. Segregating brain images into numerous non-overlapping, consistent regions having similar structure, intensity and texture is crucial in automated diagnosis. Several radiological medical image acquisition methods have been developed gradually over the years including X-ray imaging, Computed Tomography (CT), Positron Emission Tomography (PET), and Magnetic Resonance Imaging (MRI), of which MRI is of significant use in the study of the human brain, as it facilitates multi-modal tissue acquisition with diverse contrasts (T1/T2/PD-weighted), and helps in early detection of anomalies with brain tissues and organs. Owing to these advantages, researchers have tried developing several techniques over the decades for segmentation of brain MRI images with

the intent of highlighting several areas of the human brain including White Matter (WM), Gray Matter (GM) and Cerebrospinal Fluid (CSF) to aid in accurate diagnosis. The complex structure of the human brain makes region segregation a challenging task [1–6].

Image data clustering has proved itself to be a reliable approach in this use case. However, the challenge with clustering techniques is that the performance of the algorithms decreases significantly in the presence of corruptions and inhomogeneities in images. Noise corruptions including Rician Noise [7], Salt and Pepper Noise, and Gaussian Noise [8] are often found in medical images due to faulty acquisition or transmission, which degrade the image quality to the point where image processing tasks fail to get the intended results. As a result, robust techniques are required that can deliver substantial results even in the presence of noise and other aberrations.

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Prior literature presents fuzzy approaches [9–22] that account for uncertainties in class belongingness and enable data points to be assigned to several classes, each with different memberships. Spatial information negligence during pixel clustering is a significant bottleneck posed by this approach which makes it noise sensitive. A contemporary paradigm was introduced with the assimilation of rough set theory in image segmentation, such as Rough Fuzzy C Means (RFCM) [23], Rough Kernelized FCM (RKFCM) [24], Rough Spatial Kernel FCM (RSKFCM), [25], and Contextual Kernelized Rough Fuzzy C Means (KRFCMSC) [26]. Additionally, the Adaptive Rough-fuzzy Kernelized Clustering algorithm (ARKFCM)[27] and others [28–31] have further expanded this paradigm. Rough set theory helps in coping with imperceptibility, ambiguity, vagueness, and overlapping regions thus yielding decent segmentation results.

In this paper, we leverage rough and fuzzy set theories with conditional spatial tuning [32] to propose two algorithms: (1) Adaptive Rough Conditional Spatial Fuzzy C Means (ARCSFCM) and (2) Adaptive Rough **Kernelized** Conditional Spatial Fuzzy C Means (ARKCSFCM) algorithms aimed at noisy brain tissue segmentation. The presented techniques efficiently tackle uncertainties in class definitions and deal with overlapping regions and imprecise data points. Brain images are strongly correlated, and data points share common characteristics in a local neighborhood. The presented methods model the segmentation problem as a pixel classification problem where image pixels are classified into several homogeneous regions. Spatial features amalgamated with control parameters in pixel membership calculation prudently include local neighborhood influence thereby generating clusters along with compensation for corruptions and artifacts.

To conduct a comparative analysis of performance, we acquired four distinct sets of T1-weighted MRI data volumes, each comprising 71 images, sourced from the McConnell Brain Imaging Centre at the Montreal Neurological Institute [33]. To evaluate the robustness of the proposed methods, the MRI datasets were deliberately subjected to various types and intensities of noise. This controlled degradation enabled a comprehensive examination of the algorithms under challenging and realistic conditions. Performance was assessed using quantitative cluster validity indices, including the Partition Coefficient and Partition Entropy, alongside segmentation accuracy metrics. Both qualitative findings and statistical results consistently demonstrate that our approaches outperform existing state-of-the-art techniques.

The key contributions of this work are as follows:

- A pair of unsupervised segmentation algorithms is proposed that integrates rough set theory with fuzzy logic, enabling effective handling of uncertainty, overlap, and noise in medical imaging data.
- A rough set-based centroid selection mechanism is introduced to eliminate randomness in initialization, reduce redundant centroids, and significantly improve clustering stability and convergence speed.
- An adaptive membership computation approach is designed using conditional spatial parameters, which carefully incorporates local neighborhood information to enhance segmentation accuracy in the presence of noise and intensity inhomogeneity.

The rest of the paper is organized as follows: Section 2 presents the background and related work, the proposed algorithm is presented in Section 3 and 4 details the experimental results and we conclude in Section 5.

Code Metadata:

The supplementary source code and additional experimental results for the algorithms presented in this paper are publicly available at: <https://github.com/rudrajit1729/ARCSFCM-ARKCSFCM>

2. Related work

2.1. Fuzzy C-means (FCM)

Fuzzy C-Means (FCM) is an unsupervised clustering algorithm that minimizes an objective function through iterative optimization. Given an image with N pixels partitioned into c clusters, the cost function is:

$$J = \sum_{i=1}^c \sum_{k=1}^N \mu_{ki}^q \|y_i - v_k\|^2 \quad (1)$$

where μ_{ki} is the membership of pixel y_i in cluster k , v_k is the cluster centroid, and q is the fuzzifier constant. Membership and centroid values are updated as:

$$\mu_{ki} = \frac{1}{\sum_{j=1}^c \left(\frac{\|y_i - v_k\|}{\|y_i - v_j\|} \right)^{\frac{2}{q-1}}}, \quad v_k = \frac{\sum_{j=1}^N \mu_{kj}^q y_j}{\sum_{j=1}^N \mu_{kj}^q} \quad (2)$$

Although effective for noise-free data, the reliance on Euclidean distance makes FCM sensitive to image noise and intensity inhomogeneity.

2.2. Spatial fuzzy C-means (SFCM)

To improve noise resilience, Spatial FCM (SFCM) introduces a spatial neighborhood term into the objective function, enforcing local continuity:

$$J_m = \sum_{i=1}^N \sum_{k=1}^c \mu_{ki}^q \|y_i - v_k\|^2 + \frac{\alpha}{|P|} \sum_{i=1}^N \sum_{k=1}^c \mu_{ki}^q \sum_{r \in P} \|y_r - v_k\|^2 \quad (3)$$

where α controls the influence of the spatial term and P denotes the local neighborhood window. This spatial regularization significantly reduces noise sensitivity but increases computational complexity to $\mathcal{O}(N \times C \times T \times W^2)$, where W is the window size.

2.3. Kernelized FCM (KFCM)

Kernel methods enhance nonlinear separability by mapping input data into higher-dimensional feature spaces using kernel functions that replace inner products:

$$K(x, y) = e^{-\frac{\|x-y\|^2}{2\sigma^2}} \quad (4)$$

KFCM incorporates this concept into FCM, defining the kernel-based cost function as:

$$J_m = 2 \sum_{i=1}^N \sum_{k=1}^c \mu_{ki}^q (1 - K(y_i, v_k)) \quad (5)$$

The update expressions are:

$$\mu_{ki} = \frac{(1 - K(y_i, v_k))^{-\frac{1}{q-1}}}{\sum_{j=1}^c (1 - K(y_i, v_j))^{-\frac{1}{q-1}}}, \quad v_k = \frac{\sum_{i=1}^N \mu_{ki}^q y_i K(y_i, v_k)}{\sum_{i=1}^N \mu_{ki}^q K(y_i, v_k)} \quad (6)$$

KFCM provides improved nonlinear decision boundaries and better noise immunity compared to classical FCM.

2.4. Conditional spatial FCM (csFCM)

The csFCM model further refines segmentation by integrating spatial correlations among neighborhood pixels. Local membership and

centroids are computed as:

$$\mu_{ki} = \frac{1}{\|y_i - v_k\|^{\frac{2}{q-1}} \sum_{j=1}^c \mu_{kj}^q}, \quad v_k = \frac{\sum_{i=1}^N \mu_{ki}^q y_i}{\sum_{i=1}^N \mu_{ki}^q} \quad (7)$$

The conditional spatial variable is:

$$f_{ki} = \frac{\sum_{r \in P} \mu_{kr}}{|P|} \quad (8)$$

And the global membership is defined as $U_{ki} = f_{ki} \cdot \mu_{ki}$. Combining both memberships gives the weighted functions:

$$z_{ki} = \frac{\mu_{ki}^m U_{ki}^n}{\sum_{j=1}^c \mu_{kj}^m U_{kj}^n}, \quad w_k = \frac{\sum_{i=1}^N z_{ki} y_i}{\sum_{i=1}^N z_{ki}} \quad (9)$$

Here, m and n balance global and local influences. csFCM enhances robustness against noise and improves segmentation accuracy in MR images.

2.5. Rough fuzzy c-means (RFCM)

Rough Set Theory (RST) provides a framework for representing uncertain or incomplete data through lower and upper approximations. Integrating RST with FCM leads to Rough FCM (RFCM), which combines fuzzy memberships with rough approximations. The cost function is defined as:

$$J_M = \begin{cases} \omega A_L + (1 - \omega) B_L, & \text{if both } \underline{A}(\beta_i), \underline{B}(\beta_i) \neq \emptyset \\ A_L, & \text{if } \underline{B}(\beta_i) = \emptyset \\ B_L, & \text{if } \underline{A}(\beta_i) = \emptyset \end{cases} \quad (10)$$

where

$$A_L = \sum_{k=1}^c \sum_{y_i \in \underline{A}(\beta_i)} \mu_{ki}^q \|y_i - v_k\|^2, \quad B_L = \sum_{k=1}^c \sum_{y_i \in \underline{B}(\beta_i)} \mu_{ki}^q \|y_i - v_k\|^2 \quad (11)$$

and ω controls the influence of boundary regions. RFCM partitions data into lower approximation, boundary, and upper approximation regions, efficiently handling ambiguity, vagueness, and overlapping clusters.

The evolution from FCM to RFCM demonstrates progressive improvements in robustness and segmentation quality. SFCM and csFCM introduce spatial regularization, KFCM improves nonlinear separability using kernel mapping, and RFCM integrates rough approximations for handling uncertainty in complex image data.

3. Proposed methodology

In this work, we propose two adaptive rough-fuzzy conditional spatial clustering algorithms for brain MRI tissue segmentation. The methods combine rough sets to manage uncertainty and fuzzy logic to handle overlapping tissue regions. By integrating spatial neighborhood information into membership calculations, the approach improves robustness against noise and intensity inhomogeneity. Furthermore, a rough set-based centroid initialization strategy is introduced to accelerate convergence and enhance clustering stability.

3.1. Rough set-based initial cluster centroid selection

We designed an initial cluster centroid selection method to eliminate randomness in centroid initialization and improve clustering convergence. The process involves three steps: computing fuzzy membership values for pixel features, constructing a decision table to identify potential centroids, and removing redundant candidates to obtain the initial centroids.

3.1.1. Fuzzy membership computation

Each pixel of an image (dimensions $M \times N$) has n features, and each feature corresponds to each channel (represented as $F_1, F_2, \dots, F_n$). The initial set of centroids ($C_1, C_2, \dots, C_H$) is chosen, where C_i denotes the centroid pixel intensity and $2 < H < \sqrt{M \times N}$. Each of the features is allowed to belong to one or more of the three fuzzy sets (Low, Medium, High) characterized using a fuzzy membership value as defined in Eq. (12).

$$\Psi(\eta_i) = \Pi(\eta_i, \sigma, r) = \begin{cases} 2 \left(1 - \frac{(\eta_i - \sigma)}{r} \right)^2, & \text{if } \frac{r}{2} \leq |\eta_i - \sigma| \leq r \\ 1 - 2 \left(1 - \frac{(\eta_i - \sigma)}{r} \right)^2, & \text{if } 0 \leq |\eta_i - \sigma| \leq \frac{r}{2} \\ 0, & \text{Otherwise} \end{cases} \quad (12)$$

where $r(r > 0)$ is defined as the radius of the Ψ -membership function with σ as the central point. The values of r and σ can be different for each of the fuzzy sets and are chosen in such a manner that the membership functions of the fuzzy sets have an overlapping nature.

3.1.2. Constructing the decision table

Since the number of initially picked clusters (H) can fluctuate, a situation can develop where it takes a big value. A decision table is used to limit the number of clusters to acquire the optimal starting set of centroids. Each feature ($F_1, F_2, \dots, F_n$) along with its corresponding membership values ($\Psi_1, \Psi_2, \dots, \Psi_n$) is inserted in the table. Features are considered to be conditional attributes and their memberships act as decision attributes.

3.1.3. Redundant centroid elimination

After the decision table is constructed, redundant centroids are eliminated to obtain an optimal set of initial cluster centers. Two centroids C_u and C_v are considered redundant if they share identical feature sets ($F_u = F_v$), identical boundary measures ($\Psi_u = \Psi_v$), or both. In such cases, only one centroid is retained, ensuring that only distinct representatives are used during clustering. This reduction is based on data point similarity and improves the initialization process. The complete centroid selection procedure is outlined in Algorithm 1.

Algorithm 1: Proposed Algorithm: Redundant Centroid Elimination.

- 1 Select number of clusters Q , where $2 < Q < \sqrt{M \times N}$
 - 2 Compute fuzzy Ψ -membership values of image features based on Eq. (12).
 - 3 Construct decision table with each feature ($F_1, F_2, \dots, F_n$) along with their corresponding membership values ($\Psi_1, \Psi_2, \dots, \Psi_n$)
 - 4 Remove the redundant centroids which satisfy either or both of the conditions ($\eta_u = \eta_v$ or $\Psi_u = \Psi_v$ corresponding to centroids u and v)
 - 5 Return the obtained C initial centroids.
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3.2. Adaptive rough conditional spatial fuzzy C means algorithm (ARCSFCM)

After initializing the cluster centroids, we employ a rough set-based conditional spatial FCM algorithm (see Algorithm 2) for brain tissue segmentation. In this model, each cluster (ξ_j) is defined by its lower and upper approximations, denoted as $\underline{E}(\xi_j)$ and $\overline{E}(\xi_j)$, with the boundary region $B(\xi_j) = \overline{E}(\xi_j) - \underline{E}(\xi_j)$. Data points in the lower approximation belong exclusively to a single cluster and are not influenced by other clusters. In contrast, points located in the boundary region have probabilistic associations with multiple clusters and are thus subject to fuzzification. By distinguishing between these two regions, only boundary points contribute to uncertainty, ensuring more precise centroid

updates. The corresponding cost function optimized by the proposed method is presented in Eq. (13).

Algorithm 2: Adaptive Rough Set based Conditional Spatial Fuzzy C Means Algorithm.

Input: Input Image (I)

Output: Segmented Image

```

1 Set number of clusters (C), maximum iterations (MI), fuzziness
  control parameters  $m = 2, n = 2, q = 2$ , threshold  $T$  and  $\bar{T}$ .
2 Initialize the cluster centroids based on Algorithm 1.
3 Set initial membership values  $\mu^0$  is 0 and counter  $X$  as 0.
4 while  $X < MI$  do
5   Calculate membership values  $\mu_{ki}$  corresponding to all
     clusters for every pixel using cluster centroids  $v^{X-1}$  based
     on Eq. (21).
6   Calculate local membership values  $U_{ki}$  for every pixel using
     the computed global membership values  $\mu_{ki}$  as defined in
     Eq. (24).
7   The weighted membership value  $z_{ki}$  is computed for every
     pixel using the global and local membership values as
     defined in Eq. (26).
8   for every pixel in I do
9     The pixel is classified to belong to the cluster (g)
       corresponding to which maximum weighted
       membership  $z$  is obtained.
10    for  $j = 1 \dots C$  do
11      if  $|z_{ji} - z_{gi}| \leq T$  then
12         $y_i$  is assigned to upper approximation spaces  $\bar{E}(\xi_j)$ 
        and  $\bar{E}(\xi_g)$ 
13      if  $y_i \notin \text{upper approximation spaces}$  then
14        assign  $y_i$  to lower approximation space  $\underline{E}(\xi_g)$ 
15    Compute cluster centroid  $v^X$  based on Eq. (23).
16    Update the joint cluster centroids ( $H^X$ ) using Eq. (27).
17    if  $|H^X - H^{X-1}| < \bar{T}$  then
18      Stop
19     $X = X + 1$ 

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$$J_{ARCSFCM} = \begin{cases} \omega \times J_L + (1 - \omega) \times J_B, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) \neq \phi \\ J_L, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) = \phi \\ J_B, & \text{if } \underline{E}(\xi_i) = \phi \text{ and } B(\xi_i) \neq \phi \end{cases} \quad (13)$$

where

$$J_L = \sum_{k=1}^c \sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \|\bar{y}_i - v_k\|^2 + \sum_{k=1}^c \lambda_i \sum_{\bar{y}_i \in \underline{E}(\xi_i)} (1 - \mu_{ki}) \quad (14)$$

$$J_B = \sum_{k=1}^c \sum_{\bar{y}_i \in B(\xi_i)} \mu_{ki}^q \|\bar{y}_i - v_k\|^2 + \sum_{k=1}^c \lambda_i \sum_{\bar{y}_i \in B(\xi_i)} (1 - \mu_{ki}) \quad (15)$$

ω and $(1 - \omega)$ signify the relative importance assigned to the lower approximation space and the boundary zone, respectively, while λ_i serves as a scaling parameter regulating the influence of the corresponding cluster. Each image pixel's intensity, along with its local information, is regarded as a feature. To achieve this, a 3×3 window surrounding the pixel of interest (y_i) is extracted. Utilizing this neighborhood, the augmented data point ($\bar{y}_i$) is computed using Eq. (16) and employed as a data point during clustering. The incorporation of spatial features

during segmentation compensates for noise corruptions and aberrations.

$$\bar{y}_i = \frac{\sum_{x \in P_i} r_x \cdot x}{\sum_{x \in P_i} r_x} \quad (16)$$

In this context, r_x denotes the rank assigned to each pixel within its neighborhood. This rank serves as a measure of the degree of intensity deviation observed in a pixel with respect to its surrounding area. Specifically, r_x is defined as the cardinality of the set $R_x = \{z \in P_x \mid ||z - x|| < T\}$, where R_x encompasses all immediate neighbors of the pixel (x) whose intensity difference from x falls below the threshold (T). The threshold value varies depending on the input image and is determined using Eq. (17).

$$T = \frac{1}{3N} \sqrt{\frac{\pi}{2}} \sum_{i=1}^N |P_i \star W| \quad (17)$$

where

$$W = \begin{pmatrix} 2 & -5 & 2 \\ -5 & 11 & -5 \\ 2 & -5 & 2 \end{pmatrix} \quad (18)$$

and $|P_i \star W|$ represents the total intensity of the neighborhood P_i after convolving the same with W . Optimizing the cost function with respect to the membership function and the centroid under defined constraints give the necessary and sufficient conditions for the membership function and the cluster centroids to be at the minimal saddle point. Taking the partial derivative of J_L with respect to the membership function and equating it to zero we get:

$$\begin{aligned} \frac{\partial J_L}{\partial \mu_{ki}} &= 0 \\ \Rightarrow q \mu_{ki}^{q-1} \|\bar{y}_i - v_j\|^2 - \lambda_i &= 0 \\ \Rightarrow \mu_{ki} &= \left(\frac{\lambda_i}{q}\right)^{\frac{1}{q-1}} \times \left[\frac{1}{\|\bar{y}_i - v_j\|^2}\right]^{\frac{2}{q-1}} \end{aligned} \quad (19)$$

As $\sum_{q=1}^c \mu_{qp} = 1$, we get

$$\begin{aligned} \sum_{j=1}^c \left(\frac{\lambda_i}{q}\right)^{\frac{1}{q-1}} \times \left[\frac{1}{\|\bar{y}_i - v_j\|^2}\right]^{\frac{2}{q-1}} &= 1 \\ \Rightarrow \left(\frac{\lambda_i}{q}\right)^{\frac{1}{q-1}} &= \frac{1}{\sum_{j=1}^c \left(\frac{1}{\|\bar{y}_i - v_j\|^2}\right)^{\frac{2}{q-1}}} \end{aligned} \quad (20)$$

From Eqs. (19) and (20),

$$\mu_{ki} = \frac{\|\bar{y}_i - v_k\|^{-\frac{2}{q-1}}}{\sum_{j=1}^c \|\bar{y}_i - v_j\|^{-\frac{2}{q-1}}} \quad (21)$$

On equating the partial derivative of J_L w.r.t the centroid to zero we get:

$$\begin{aligned} \frac{\partial J_L}{\partial v_k} &= 0 \\ \Rightarrow \sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \times 2 \|\bar{y}_i - v_k\| \times (-1) &= 0 \\ \Rightarrow v_k \sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q &= \sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \bar{y}_i \\ \Rightarrow v_k &= \frac{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q} \end{aligned} \quad (22)$$

The partial derivatives of J_B yield similar results. Finally, the centroids are computed using Eq. (23).

$$v_k = \begin{cases} \omega \frac{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q} + (1 - \omega) \frac{\sum_{\bar{y}_i \in B(\xi_i)} \mu_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in B(\xi_i)} \mu_{ki}^q}, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) \neq \phi \\ \frac{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} \mu_{ki}^q}, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) = \phi \\ \frac{\sum_{\bar{y}_i \in B(\xi_i)} \mu_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in B(\xi_i)} \mu_{ki}^q}, & \text{if } \underline{E}(\xi_i) = \phi \text{ and } B(\xi_i) \neq \phi \end{cases} \quad (23)$$

Furthermore, the spatial aspect of the algorithm based on conventional csFCM is explored by introducing a conditional spatial membership U_{ki} corresponding to the belongingness of the data point $\bar{y}_i$ in the cluster ξ_i as defined to Eq. (24).

$$U_{ki} = CP_{ki} \times \mu_{ki} \quad (24)$$

where

$$CP_{ki} = \frac{\sum_{r \in P} \mu_{kr}}{|P|} \quad (25)$$

where CP_{ki} is the conditional parameter which decides the involvement level of the concerned data point in the k^{th} cluster by including the spatial neighborhood details around the concerned pixel. Here P is the square neighborhood window centered at the pixel y_i and $|P|$ is its cardinality. In this paper, a 3×3 window is extracted for this purpose as larger neighborhood over-smooths the edges resulting in an increase in pixel misclassification. On combining the local and global membership functions, a weighted membership function is computed which incorporates the global features along with the local spatial information as defined in Eq. (26).

$$z_{ki} = \frac{(U_{ki})^m \times (\mu_{ki})^n}{\sum_{j=1}^c (U_{ji})^m \times (\mu_{ji})^n} \quad (26)$$

In the proposed method, the parameters m and n (set to 2) control the balance between global and local membership components in the weighted membership function. These parameters help regulate sensitivity to minor membership variations, ensuring stable clustering. When pixel correlations are strong, as is typical in medical images, the method increases the likelihood of assigning a data point to the same cluster as its neighbors. In homogeneous regions, local membership has minimal influence, while in noisy or intensity-inhomogeneous regions, spatial context guides more accurate labeling. By integrating rough set principles, a joint cluster centroid is computed to handle vague, incomplete, or ambiguous class boundaries, as defined by Eq. (27).

$$H_k = \begin{cases} \omega \frac{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} z_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} z_{ki}^q} + (1 - \omega) \frac{\sum_{\bar{y}_i \in B(\xi_i)} z_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in B(\xi_i)} z_{ki}^q}, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) \neq \phi \\ \frac{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} z_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in \underline{E}(\xi_i)} z_{ki}^q}, & \text{if } \underline{E}(\xi_i) \neq \phi \text{ and } B(\xi_i) = \phi \\ \frac{\sum_{\bar{y}_i \in B(\xi_i)} z_{ki}^q \bar{y}_i}{\sum_{\bar{y}_i \in B(\xi_i)} z_{ki}^q}, & \text{if } \underline{E}(\xi_i) = \phi \text{ and } B(\xi_i) \neq \phi \end{cases} \quad (27)$$

3.3. Adaptive rough kernelized conditional spatial fuzzy C means algorithm (ARKCSFCM)

This section presents a rough-fuzzy adaptive kernel-based conditional spatial clustering algorithm (see Algorithm 3) for brain MRI tissue segmentation. Following rough set principles, points in the lower approximation are treated as definite cluster members, unaffected by other clusters, while boundary points carry probabilistic association and do not influence centroid updates. Only boundary points are fuzzified, and the objective function is optimized as defined in Eq. (28).

$$J_{ARKCSFCM} = \begin{cases} \psi \times J_L + (1 - \psi) \times J_B, & \text{if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) \neq \Phi \\ J_L, & \text{if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) = \Phi \\ J_B, & \text{if } \underline{R}(C_i) = \Phi \text{ and } B(C_i) \neq \Phi \end{cases} \quad (28)$$

where

$$J_L = \sum_{k=1}^c \sum_{\bar{y}_i \in \underline{R}(C_i)} \mu_{ki}^m (1 - K(\bar{y}_i, v_k)) + \sum_{k=1}^c \phi_i \sum_{\bar{y}_i \in \underline{R}(C_i)} (1 - \mu_{ki}) \quad (29)$$

Algorithm 3: Adaptive Rough Set based Kernelized Conditional Spatial Fuzzy C Means Algorithm.

Input: Input Image (I)
Output: Segmented Image

- 1 Set number of clusters (C), maximum iterations (MI), fuzziness control parameters $m = 2, n = 2, q = 2$, threshold T and $\bar{T}$.
- 2 Initialize the cluster centroids based on Algorithm 1.
- 3 Set initial membership values μ^0 is 0 and counter X as 0.
- 4 **while** $X < MI$ **do**
- 5 Calculate membership values μ_{ki} corresponding to all clusters for every pixel using cluster centroids v^{X-1} based on Eq. (33).
- 6 Calculate local membership values U_{ki} for every pixel using the computed global membership values μ_{ki} as defined in Eq. (36).
- 7 The weighted membership value Z_{ki} is computed for every pixel using the global and local membership values as defined in Eq. (38).
- 8 **for every pixel in I do**
- 9 The pixel is classified to belong to the cluster (g) corresponding to which maximum weighted membership z is obtained.
- 10 **for** $j = 1 \dots C$ **do**
- 11 **if** $|Z_{ji} - Z_{gi}| \leq T$ **then**
- 12 y_i is assigned to upper approximation spaces $\bar{E}(\xi_j)$ and $\bar{E}(\xi_g)$
- 13 **if** $y_i \notin \text{upper approximation spaces}$ **then**
- 14 assign y_i to lower approximation space $\underline{E}(\xi_g)$
- 15 Compute cluster centroid v^X based on Eq. (35).
- 16 Update the joint cluster centroids (H^X) using Eq. (39).
- 17 **if** $|H^X - H^{X-1}| < \bar{T}$ **then**
- 18 Stop
- 19 $X = X + 1$

$$J_B = \sum_{k=1}^c \sum_{\bar{y}_i \in B(C_i)} \mu_{ki}^m (1 - K(\bar{y}_i, v_k)) + \sum_{k=1}^c \phi_i \sum_{\bar{y}_i \in B(C_i)} (1 - \mu_{ki}) \quad (30)$$

$(1 - \psi)$ controls the influence of the boundary region and ϕ_i is a scale tuner which controls the influence of the cluster. $K(\bar{y}_i, v_k)$ represents a circular kernel as defined in Eq. (31). It helps speed up the convergence of the algorithm and boosts the performance of the presented technique.

$$K(\bar{y}_i, v_k) = \frac{2}{\pi} \cos^{-1} \left(\frac{-\|\bar{y}_i - v_k\|}{\sigma} \right) - \frac{2}{\pi} \frac{\|\bar{y}_i - v_k\|}{\sigma} \sqrt{1 - \left(\frac{\|\bar{y}_i - v_k\|}{\sigma} \right)^2} \quad (31)$$

σ serves as the tuning parameter for adjusting the kernel.

The pixel intensity value along with its neighborhood information is considered a data point. For this, firstly a 3×3 neighborhood (P) centered at the concerned pixel (y_i) is extracted. Using the neighborhood the augmented data point $\bar{y}_i$ is calculated using Eq. (16) and used as a data point during clustering. The noise corruptions and aberrations are compensated for owing to the spatial feature consideration during segmentation. Optimizing the objective function with respect to the membership function and the centroids respectively under the specified constraints we get:

$$\frac{\delta J_L}{\delta \mu_{ki}} = 0, \quad \frac{\delta J_L}{\delta v_k} = 0 \quad (32)$$

After solving we obtained:

$$\mu_{ki} = \frac{(1 - K(\bar{y}_i, v_k))^{-\frac{1}{m-1}}}{\sum_{j=1}^c (1 - K(\bar{y}_j, v_k))^{-\frac{1}{m-1}}} \quad (33)$$

$$v_k = \frac{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m K(\bar{y}_i, v_k)} \quad (34)$$

Similar results are obtained for J_B . By combining the results for J_L and J_B , the centroid computation function becomes:

$$v_k = \begin{cases} \phi \frac{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m K(\bar{y}_i, v_k)} \\ + (1 - \phi) \frac{\sum_{\bar{y}_i \in B(C_i)} \mu_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in B(C_i)} \mu_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) \neq \Phi \\ \frac{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in R(C_i)} \mu_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) = \Phi \\ \frac{\sum_{\bar{y}_i \in B(C_i)} \mu_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in B(C_i)} \mu_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) = \Phi \text{ and } B(C_i) \neq \Phi \end{cases} \quad (35)$$

Now, the conditional spatial aspect of the technique is unwrapped by introducing a conditional spatial membership function U_{ki} which corresponds to the data point $\bar{y}_i$, and the cluster C_k .

$$U_{ki} = \mu_{ki} \times \xi_{ki} \quad (36)$$

where

$$\xi_{ki} = \frac{\sum_{s \in N} \mu_{ks}}{|N|} \quad (37)$$

ξ_{ki} acts as a conditional tuner that determines whether a data point contributes to the k^{th} cluster centroid based on its spatial context. A 3×3 neighborhood (N) around pixel y_i is used to avoid edge smoothing and misclassification. The final weighted membership function combines global and local memberships, integrating both pixel features and neighborhood information. It includes the overall features along with neighborhood details and is defined as follows:

$$Z_{ki} = \frac{(U_{ki})^a \times (\mu_{ki})^b}{\sum_{j=1}^c (U_{ji})^a \times (\mu_{ji})^b} \quad (38)$$

Parameters a and b control the balance between global and local membership components in the weighted function. A joint cluster centroid function is defined in Eq. (39) which is formulated using the weighted membership and is used to compute cluster centroids after each iteration of the algorithm. The proposed rough fuzzy set based kernelized conditional spatial C means clustering algorithm is summarized in Algorithm 3.

$$H_k = \begin{cases} \phi \frac{\sum_{\bar{y}_i \in R(C_i)} Z_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in R(C_i)} Z_{ki}^m K(\bar{y}_i, v_k)} \\ + (1 - \phi) \frac{\sum_{\bar{y}_i \in B(C_i)} Z_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in B(C_i)} Z_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) \neq \Phi \\ \frac{\sum_{\bar{y}_i \in R(C_i)} Z_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in R(C_i)} Z_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) \neq \Phi \text{ and } B(C_i) = \Phi \\ \frac{\sum_{\bar{y}_i \in B(C_i)} Z_{ki}^m \bar{y}_i K(\bar{y}_i, v_k)}{\sum_{\bar{y}_i \in B(C_i)} Z_{ki}^m K(\bar{y}_i, v_k)}, \text{ if } \underline{R}(C_i) = \Phi \text{ and } B(C_i) \neq \Phi \end{cases} \quad (39)$$

4. Experimental results

This section reports the experimental results used to test the robustness of the proposed techniques against the existing state-of-the-art methods. For this purpose, data sets have been obtained from the McConnell Brain Imaging Centre of the Montreal Neurological Institute, McGill University [33]. We have collected different T1 and T2-weighted data volumes consisting of 71 images each (slices 50 to 121). Each of these images has the Gray Matter (GM), White Matter (WM), and Cerebrospinal Fluid (CSF) regions. In this paper, we report experimental observations and inferences from algorithmic performances using two MR image data volumes corrupted by 9% and 11% noise respectively and accompanied by 40% inhomogeneity (IIH) each. The rationale for introducing IIH is to test the performance of algorithms in the presence of heterogeneity in image data. The IIH value is fixed at 40% as this work is more interested in studying the effect of different techniques in the context of varying noise corruptions in brain images. The collected image size is $1 \text{ mm} \times 1 \text{ mm} \times 1 \text{ mm}$ and the image resolution is $181 \times 217 \times 181$ voxels.

4.1. Quantitative metrics

For quantitative evaluation, cluster validity functions (Partition coefficient (V_{pc}) and Partition entropy (V_{pe})) and tissue segmentation accuracy metrics (TSA and F1-score) are utilized. Cluster validity functions are essential indicators of fuzzy partition and algorithmic robustness is highlighted when the V_{pc} and V_{pe} values tend to 0 and 1 respectively (i.e., higher V_{pc} values and lower V_{pe} values are indicators of robust segmentation results). The equations for V_{pc} , V_{pe} , TSA , and F1-score are given below (Eqs. 40–43):

$$V_{pc} = \frac{1}{n} \sum_{k=1}^C \sum_{i=1}^n \mu_{ki}^2, \quad V_{pe} = \frac{1}{n} \sum_{k=1}^C \sum_{i=1}^n -\mu_{ki} \log(\mu_{ki}) \quad (40)$$

$$TSA = \frac{(TP + TN)}{(TP + FP + TN + FN)} \quad (41)$$

$$Precision = \frac{TP}{(TP + FP)}, \quad Recall = \frac{TP}{(TP + FN)} \quad (42)$$

$$F1\text{-score} = \frac{(2 \times Precision \times Recall)}{(Precision + Recall)} \quad (43)$$

where TP , TN , FP , and FN represent the true positive, true negative, false positive, and false negative. TSA is an important metric that judges the efficiency of the clustering algorithm in tissue segmentation. High TSA and F1-score values indicate minimum deviation from ground truth images thus signifying robust tissue segregation.

4.2. Performance evaluation

The proposed ARCSFCM and ARKCSFCM algorithms are compared against ten standard fuzzy clustering baselines (vFCM, FLICM, ARFCM, NLFCM, RFCM, KRFCMSC, sRFCM, ARKFCM, EKCSFCM, IMKFCM) on simulated BrainWeb T1- and T2-weighted MRI data corrupted with 9% and 11% noise and 40% intensity inhomogeneity (IIH). For the sake of space constraints, we report result related to slice 95 in the paper. Results for the other slices are provided in the supplement.

4.2.1. Qualitative evaluation

Figs. 1–4 illustrate the segmentation results for the T1-weighted slice 95 (9% noise, 40% IIH), showing the composite segmentation map together with the extracted CSF, GM, and WM tissue maps.

In the composite map (Fig. 1), baseline methods suffer from visible noise-induced misclassifications and blurred boundaries, especially around the ventricles and cortical folds. vFCM and FLICM produce patchy GM–WM transitions and residual salt-and-pepper artifacts inside tissue regions. ARFCM and NLFCM reduce inhomogeneity but still leak intensity gradients into neighbouring classes. The kernel-based and

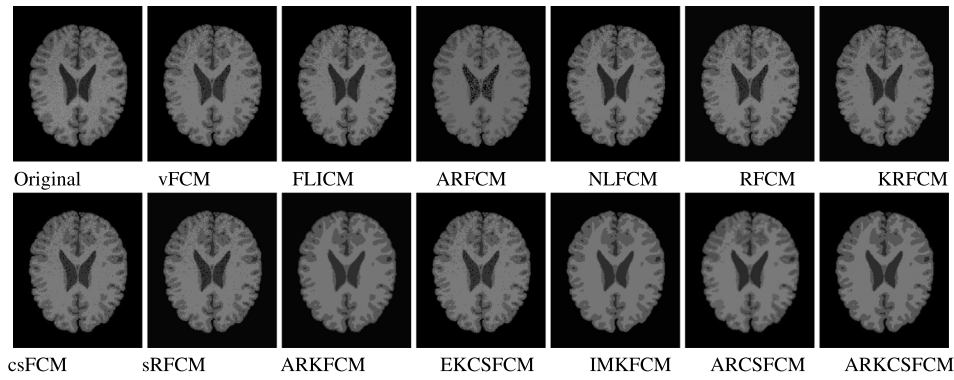


Fig. 1. Qualitative comparison of segmented images of T1-weighted brain MRI image of slice 95 with 9% noise and 40% inhomogeneity.

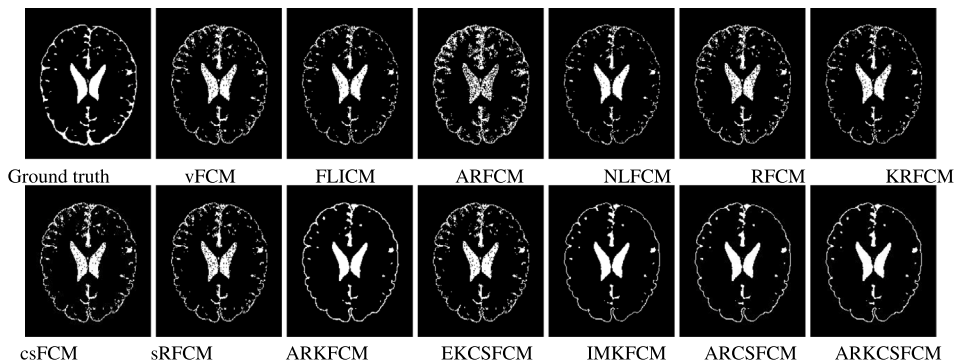


Fig. 2. Qualitative comparison for CSF images of T1-weighted brain MRI image of slice 95 having 9% noise and 40% inhomogeneity.

spatially regularized variants (RFCM, KRFCMSC, ARKFCM) improve homogeneity yet fail to recover fine details lost to noise. In sharp contrast, both ARCSFCM and ARKCSFCM deliver artefact-free boundaries that follow the ground-truth contours. The ventricular CSF appears uniformly dark with no stray GM pixels, the cortical GM ribbon is continuous without fragmentation, and the deep white-matter tracts remain sharply delineated.

The tissue-specific maps reinforce this observation. For CSF (Fig. 2), the proposed methods eliminate the false-positive bright spots that affect FLICM and NLFCM within the brain parenchyma. For GM (Fig. 3), ARKCSFCM accurately recovers the very fine outer brain layer and important regions inside the brain, detecting them correctly, whereas ARKFCM and EKCSFCM show partial-volume blurring at the GM-WM interface. For WM (Fig. 4), only the proposed algorithms are able to preserve the internal capsule and corpus callosum as compact, high-intensity structures without the fragmentation seen in IMKFCM.

These visual gains arise directly from the algorithm design. The adaptive window selects an optimal local neighborhood size that balances noise suppression with edge preservation. The rough-set lower and upper approximations explicitly model uncertainty in boundary pixels. The kernelized conditional spatial function incorporates neighborhood correlation and prevents isolated noisy pixels from influencing cluster assignment.

Similar improvements are observed for the T1-weighted slice 90 (11% noise, 40% IIH) and T2-weighted slice 100 (9% noise, 40% IIH). In both cases the proposed ARCSFCM and ARKCSFCM algorithms are able to eliminate the noise-induced misclassifications and boundary leakage still visible in other related methods, confirming robustness across noise levels and inhomogeneities.

4.2.2. Quantitative evaluation

Clustering quality is measured by the Partition Coefficient V_{pc} (higher is better) and Partition Entropy V_{pe} (lower is better). Tables 1 and 2

report these metrics for T1-weighted images with 9% noise across slices 50–120.

The near-unity V_{pc} and sub-0.01 V_{pe} of ARKCSFCM quantify extremely low fuzziness and uncertainty. The 1–2% absolute improvement over the strongest baseline translates to a 30–50% relative reduction in partition entropy, due to the rough-set boundary handling that confines ambiguous pixels to a controlled upper approximation rather than allowing them to diffuse their membership across clusters.

Identical trends hold for T1-weighted MR images with 11% noise and for T2-weighted images with 9% noise and 40% inhomogeneity. In both conditions ARKCSFCM maintains $V_{pc} > 0.990$ and $V_{pe} < 0.015$ across all slices 50–120, outperforming the strongest baseline by 1–2% in V_{pc} and achieving a 30–50% relative reduction in partition entropy.

Tissue segmentation accuracy is further quantified by Total Segmentation Accuracy (TSA) and F1-score (Table 3). For the representative T1-weighted slice 95 (9% noise), ARKCSFCM achieves TSA = 0.9704 (CSF), 0.9621 (GM), 0.9784 (WM) and corresponding F1-scores of 0.7547, 0.8642, 0.9530. These figures exceed the best baseline by 3–7 percentage points in TSA and 8–15% in F1-score.

The quantitative performance of the segmentation algorithms is evaluated by averaging the Accuracy and F1-score across different slices and three tissue regions (CSF, GM, and WM). The averaged results are summarized in Table 4.

From the results, it is evident that segmentation performance varies across algorithms and tissue types. Conventional methods such as vFCM, RFCM, and ARFCM exhibit comparatively lower mean Accuracy and F1-score values. In particular, ARFCM shows a noticeable degradation in F1-score, especially in the CSF region, indicating poor boundary delineation and sensitivity to noise. Similarly, csFCM and sRFCM demonstrate moderate performance but lack consistency across different tissues, as reflected by their lower mean F1-scores.

Improved variants such as FLICM, NLFCM, and KRFCMSC show enhanced performance due to their ability to incorporate spatial

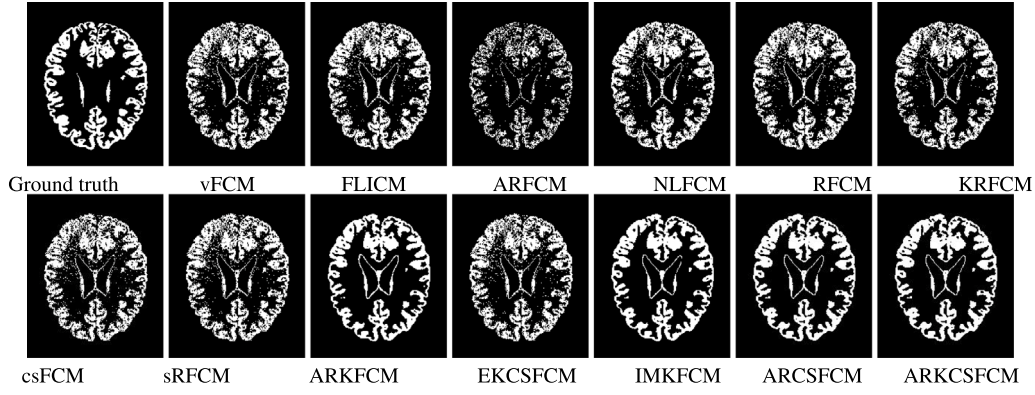


Fig. 3. Qualitative comparison for GM images of T1-weighted brain MRI image of slice 95 having 9% noise and 40% inhomogeneity.

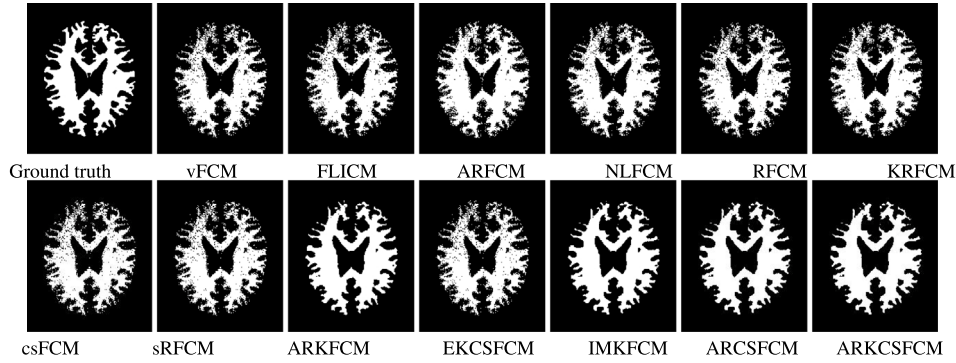


Fig. 4. Qualitative comparison of WM images of T1-weighted brain MRI image of slice 95 having 9% noise and 40% inhomogeneity.

Table 1

Comparative evaluation of V_{pc} values of T1-weighted MR images having 9% noise and 40% inhomogeneity.

Algorithm	Slice numbers								
	50	60	70	80	90	95	100	110	120
vFCM [19]	0.9345	0.7621	0.9574	0.7621	0.7412	0.8254	0.9417	0.9321	0.9417
FLICM [34]	0.90183	0.88497	0.88614	0.88898	0.90218	0.91056	0.91251	0.91428	0.91704
ARFCM [20]	0.92244	0.91746	0.91720	0.89701	0.91978	0.91991	0.92423	0.92990	0.92601
NLFCM [35]	0.93882	0.94672	0.92734	0.93039	0.93283	0.93803	0.93919	0.94498	0.94169
RFCM [23]	0.91620	0.90297	0.90249	0.90265	0.90917	0.92003	0.92147	0.91603	0.93021
KRFCMSC [26]	0.93559	0.92215	0.92435	0.92446	0.93491	0.94198	0.94352	0.94343	0.94614
csFCM [32]	0.9836	0.9811	0.9807	0.9818	0.9838	0.986	0.9854	0.985	0.9877
sRFCM [16]	0.91617	0.90293	0.90266	0.90271	0.91307	0.92189	0.91903	0.92178	0.93689
ARKFCM [27]	0.95828	0.95174	0.95277	0.95324	0.96005	0.96390	0.96493	0.96413	0.96632
EKCSFCM [36]	0.98355	0.98100	0.98062	0.98174	0.98370	0.98593	0.98519	0.98495	0.98760
IMKFCM [37]	0.98121	0.97886	0.97954	0.97903	0.98197	0.98329	0.98304	0.98259	0.98529
ARCSFCM	0.99089	0.98916	0.98838	0.98925	0.99096	0.99117	0.99101	0.99142	0.99162
ARKCSFCM	0.99280	0.99200	0.99430	0.99232	0.99358	0.99410	0.99428	0.99415	0.99417

information and suppress noise. Among them, NLFCM consistently achieves higher Accuracy and F1-score across all regions, demonstrating improved segmentation capability. However, slight variations in GM and WM results indicate that these methods still face challenges in handling complex tissue boundaries and intensity inhomogeneity.

The kernel-based and hybrid approaches, including ARKFCM and IMKFCM, further improve segmentation performance. These methods achieve higher mean values, particularly in WM segmentation, due to their ability to model nonlinear relationships in image data. Nevertheless, minor inconsistencies remain in CSF segmentation, suggesting sensitivity to low-contrast regions.

The proposed methods, ARCSFCM and ARKCSFCM, consistently outperform all existing approaches across all evaluation metrics.

Specifically, ARKCSFCM achieves the highest average Accuracy (0.9701, 0.9588, and 0.9763) and F1-score (0.7643, 0.8667, and 0.9499) for CSF, GM, and WM regions, respectively. This improvement can be attributed to the effective integration of adaptive regularization and kernel-based clustering, which enhances both local and global feature representation. Similarly, ARCSFCM demonstrates competitive performance, particularly in GM and WM regions, indicating its robustness in handling tissue heterogeneity.

To further validate the observed performance differences, statistical significance tests are conducted. The segmentation performance of the considered algorithms is quantitatively assessed using Accuracy and F1-score across multiple slices and tissue regions (CSF, GM, and WM). To provide a comprehensive evaluation, the results were aggregated and

Table 2Comparative evaluation of V_{pe} values of T1-weighted MR images with 9% noise and 40% inhomogeneity.

Algorithm	Slice numbers								
	50	60	70	80	90	95	100	110	120
vFCM	0.2841	0.1157	0.2952	0.1157	0.0991	0.1072	0.2340	0.3075	0.2340
FLICM	0.18858	0.22036	0.21792	0.21334	0.18849	0.17243	0.16916	0.16521	0.15861
ARFCM	0.14885	0.15991	0.16018	0.19758	0.15499	0.15409	0.14604	0.13478	0.14101
NLFCM	0.11502	0.09813	0.13587	0.12998	0.12519	0.11541	0.11334	0.10281	0.10887
RFCM	0.14970	0.17317	0.17384	0.17363	0.16223	0.14322	0.14084	0.15416	0.12813
KRFCMSC	0.11308	0.13616	0.13185	0.13235	0.11403	0.10147	0.09906	0.09912	0.09444
csFCM	0.0269	0.0312	0.0317	0.03	0.0267	0.0231	0.0242	0.0246	0.0203
sRFCM	0.14969	0.17319	0.17301	0.17360	0.15524	0.13985	0.14479	0.13929	0.11264
ARKFCM	0.06895	0.07942	0.07756	0.07722	0.06581	0.05932	0.05794	0.05913	0.05569
EKCSFCM	0.02699	0.03133	0.03188	0.03012	0.02684	0.02324	0.02448	0.02477	0.02041
IMKFCM	0.03151	0.03519	0.03391	0.03494	0.03002	0.02784	0.02834	0.02911	0.02462
ARCSFCM	0.01376	0.01633	0.01764	0.01622	0.01358	0.01325	0.01364	0.01304	0.01265
ARKCSFCM	0.0110	0.0121	0.0084	0.0116	0.0096	0.0088	0.0085	0.0088	0.0088

Table 3

Tissue segmentation accuracy (TSA) and F1-scores values for CSF, GM, and WM tissues on T1-weighted MRI brain image with 9% noise and 40% inhomogeneity (slice number 95) using different algorithms.

Regions	Slice number 95					
	CSF		GM		WM	
	Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score
vFCM	0.96038	0.72184	0.93136	0.79905	0.95519	0.90264
FLICM	0.96293	0.73216	0.93856	0.82272	0.95677	0.90721
ARFCM	0.92991	0.55547	0.91313	0.68674	0.95338	0.90669
NLFCM	0.96412	0.73905	0.94081	0.82653	0.95725	0.91030
RFCM	0.96051	0.71640	0.93240	0.80323	0.95519	0.90264
KRFCMSC	0.96111	0.75144	0.93918	0.81545	0.95656	0.91005
csFCM	0.96227	0.73239	0.93240	0.80323	0.95519	0.90264
sRFCM	0.96212	0.72352	0.93385	0.80869	0.95519	0.90264
ARKFCM	0.96092	0.78694	0.95211	0.85638	0.97182	0.94123
EKCSFCM	0.96227	0.73239	0.93240	0.80323	0.95519	0.90264
IMKFCM	0.96138	0.79406	0.95030	0.85526	0.97115	0.93849
ARCSFCM	0.96596	0.74333	0.95216	0.85865	0.97207	0.94106
ARKCSFCM	0.97041	0.75472	0.96212	0.86420	0.97841	0.95302

Table 4

Average tissue segmentation accuracy (TSA) and F1-scores of CSF, GM, and WM across slices using different algorithms.

Regions	Average over slices					
	CSF		GM		WM	
	Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score
vFCM	0.9550	0.7192	0.9221	0.7786	0.9509	0.8916
FLICM	0.9623	0.7426	0.9356	0.8191	0.9547	0.9026
ARFCM	0.9320	0.6104	0.9081	0.6746	0.9498	0.9007
NLFCM	0.9638	0.7449	0.9383	0.8236	0.9556	0.9070
RFCM	0.9576	0.7255	0.9263	0.7937	0.9518	0.8938
KRFCMSC	0.9568	0.7505	0.9359	0.8121	0.9554	0.9069
csFCM	0.9585	0.6901	0.9264	0.7908	0.9526	0.8964
sRFCM	0.9582	0.7208	0.9274	0.7945	0.9526	0.8964
ARKFCM	0.9632	0.7787	0.9501	0.8542	0.9712	0.9397
EKCSFCM	0.9576	0.7291	0.9258	0.7884	0.9526	0.8964
IMKFCM	0.9632	0.7834	0.9467	0.8481	0.9698	0.9354
ARCSFCM	0.9653	0.7361	0.9513	0.8601	0.9712	0.9391
ARKCSFCM	0.9701	0.7643	0.9588	0.8667	0.9763	0.9499

summarized in terms of mean and standard deviation, as presented in Table 5. The mean values reflect the overall segmentation accuracy, while the standard deviation indicates the stability and consistency of each method under varying image conditions.

From Table 5, it is evident that conventional clustering approaches such as vFCM, RFCM, and ARFCM achieve relatively lower mean accuracy and F1-score, accompanied by higher standard deviation values. In particular, ARFCM exhibits the largest variability (Accuracy: 0.9299 ±

Table 5

Statistical comparison of segmentation algorithms using mean ± standard deviation across all slices and regions.

Algorithm	Accuracy (Mean ± SD)	F1-score (Mean ± SD)
vFCM	0.9427 ± 0.0138	0.7965 ± 0.0842
FLICM	0.9509 ± 0.0116	0.8214 ± 0.0663
ARFCM	0.9299 ± 0.0167	0.7286 ± 0.1415
NLFCM	0.9526 ± 0.0111	0.8252 ± 0.0658
RFCM	0.9452 ± 0.0134	0.8043 ± 0.0759
KRFCMSC	0.9494 ± 0.0121	0.8232 ± 0.0669
csFCM	0.9458 ± 0.0128	0.7924 ± 0.0806
sRFCM	0.9461 ± 0.0126	0.8039 ± 0.0748
ARKFCM	0.9615 ± 0.0082	0.8575 ± 0.0517
EKCSFCM	0.9453 ± 0.0132	0.8046 ± 0.0754
IMKFCM	0.9599 ± 0.0087	0.8556 ± 0.0528
ARCSFCM	0.9626 ± 0.0079	0.8451 ± 0.0503
ARKCSFCM	0.9684 ± 0.0068	0.8603 ± 0.0475

0.0167, F1-score: 0.7286 ± 0.1415), indicating unstable segmentation performance, especially in challenging regions with low contrast and noise. Similarly, csFCM and sRFCM demonstrate moderate mean performance but with noticeable variability, suggesting limited robustness across different tissue structures.

Spatially enhanced methods, including FLICM, NLFCM, and KRFCMSC, show improved mean performance with reduced variability compared to conventional approaches. Among them, NLFCM achieves relatively higher accuracy (0.9526 ± 0.0111) and F1-score (0.8252 ± 0.0658), indicating better noise resilience and utilization of neighborhood information. However, the remaining standard deviation values suggest that these methods still experience performance fluctuations when dealing with complex anatomical variations and intensity inhomogeneity.

Kernel-based and hybrid methods, such as ARKFCM and IMKFCM, further enhance segmentation quality by modeling nonlinear data distributions. These approaches achieve higher mean accuracy (ARKFCM: 0.9615 ± 0.0082, IMKFCM: 0.9599 ± 0.0087) and F1-score, along with lower standard deviation, indicating improved robustness.

The proposed methods, ARCSFCM and ARKCSFCM, consistently outperform all existing techniques in both mean performance and stability. ARCSFCM achieves high Accuracy and F1-score (0.9626 ± 0.0079, 0.8451 ± 0.0503), while ARKCSFCM attains the best overall performance with the highest mean Accuracy and F1-score (0.9684 ± 0.0068, 0.8603 ± 0.0475) and the lowest standard deviation among all methods. The reduced variability indicates that the proposed approaches provide stable and reliable segmentation results across different slices and tissue types. This superior performance can be attributed to the integration of adaptive regularization with kernel-based clustering, which effectively captures both local structural details and global contextual relationships.

To statistically validate these observations, a one-way ANOVA test was conducted on the aggregated results. The analysis yielded a statistically significant difference among the algorithms ($p < 0.05$), confirming that the observed performance improvements are not due to random variations. The statistical significance aligns with the mean and standard deviation analysis, where methods with higher mean performance and lower variability correspond to better segmentation outcomes.

Overall, the combined quantitative and statistical evaluation demonstrates that the proposed methods not only achieve superior segmentation accuracy but also ensure consistent performance across diverse imaging conditions. This highlights their effectiveness and suitability for practical brain MRI segmentation applications and downstream volumetric or morphometric studies.

5. Conclusion

This paper presents two Adaptive Rough Fuzzy Conditional Spatial Clustering algorithms designed for robust brain tissue segmentation in MRI scans. These methods integrate rough set theory, fuzzy logic, and spatial context to enhance segmentation reliability under noisy conditions. By modeling spatial relationships and intensity variations, they effectively distinguish tissue boundaries while maintaining computational efficiency, making them suitable for field use.

A key strength of the algorithm is its resilience to artifacts, enabling precise delineation of CSF, GM, and WM regions in degraded MRI data. Its adaptive design ensures consistent performance, even in challenging imaging environments.

Future work should focus on scalability and integration into medical software systems for automated detection of brain abnormalities. Ultimately, this approach advances MRI segmentation technology, supporting earlier diagnosis and improved treatment planning in neurological applications.

CRedit authorship contribution statement

Amiya Halder: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Rudrajit Choudhuri:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Apurba Sarkar:** Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is publicly available.

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